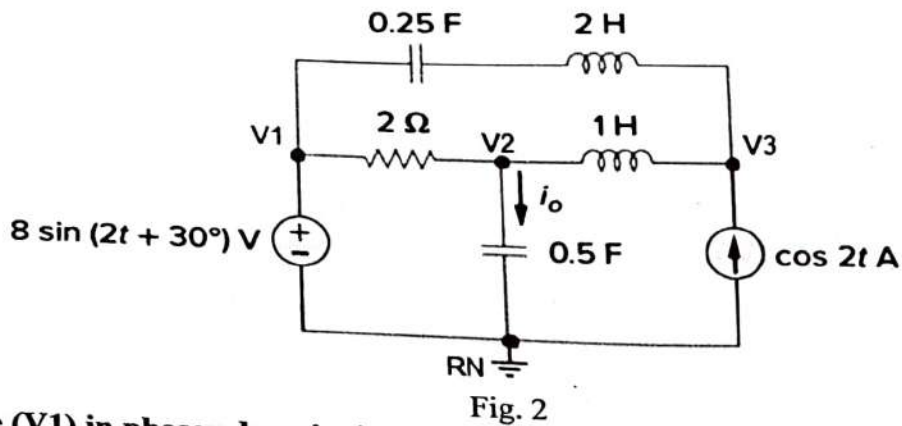


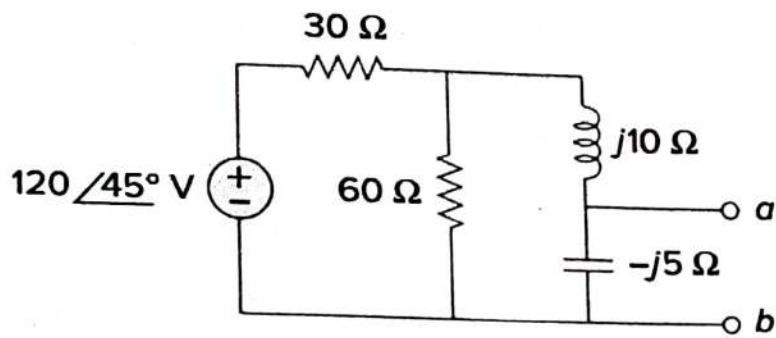
Problem 3. For the questions 10,11,12 consider the circuit shown in figure 2 and answer using nodal analysis:



10) The voltage (V_1) in phasor domain is:

11) Write the required nodal equations (do not simplify):

Problem 4. For the questions 13,14,15 consider the circuit shown in figure 3 and answer:



12) The Thevenin impedance (Z_{Th}) with respect to the terminals a-b is:

13) The Thevenin voltage (V_{Th}) with respect to the terminals a-b is:



Problem 1. For the questions 1,2,3,4,5,6 consider:

- $v(t) = 500 \cos(1000 t + 30^\circ)$ and $i(t) = -20 \sin(1000 t - 120^\circ)$.

1) The voltage leads the current in degrees by:

2) The current lags the voltage in radians by:

3) The voltage value (v) at $t=1\text{ms}$ is:

4) The phasor domain representation of $i(t)$ is:

5) The phasor domain representation of $v(t)$ is:

6) The frequency of the voltage in Hz is:

Problem 2. For the questions 7,8,9 consider the circuit shown in figure 1 and answer:

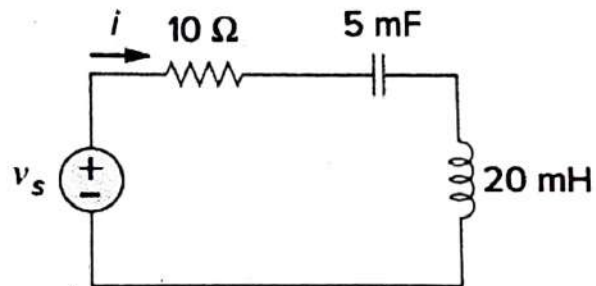


Fig.1

- $v_s(t) = 100 \cos(200 t + 45^\circ) \text{ V}$

7) The equivalent impedance at the terminals of the voltage source in (Ω) is:

8) The admittance at the terminals of the voltage source in (S) is:

9) The steady state current (i) in time domain is:

